

6CS012 – Artificial Intelligence and Machine Learning.

Lecture – 01

Foundational **Math Skills** for AI and ML.

A quick revision on **Linear Algebra** and **Derivative**.

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Learning Outcomes!!!

- Review and revise some fundamental concepts from Mathematics – **Linear Algebra and Derivative** we will be using through out the course.
 - ***Cautions !!!***
 - We will omit many important topics in Linear algebra and Matrix Calculus, which we believe are not essential for understanding deep learning.
- In Particular we will discuss:
 - **Why do we need Linear Algebra for Machine/Deep Learning?**
 - **{Almost}** Everything we need to know about vector and matrices for Machine/Deep Learning.
 - **A very big picture on Definition of Derivative and Matrix Calculus.**

A. Why do we need Linear Algebra for ML/DL?

{Why to study **Vector** and **Matrices**}

A.1 What is Linear Algebra?

- **Linear Algebra** is the branch of **mathematics** concerning **linear equations** such as:
 - $\mathbf{a}_1\mathbf{x}_1 + \dots + \mathbf{a}_n\mathbf{x}_n = \mathbf{b}$;
 - **linear maps** such as:
 - $(\mathbf{x}_1, \dots, \mathbf{x}_n) \mapsto \mathbf{a}_1\mathbf{x}_1 + \dots + \mathbf{a}_n\mathbf{x}_n$;
 - and **their representations** in **vector spaces and through matrices**. – Wikipedia.
- **Linear algebra** is a branch of mathematics that deals with **vectors, vector spaces** (also **known as linear spaces**),
 - and **linear transformations** between these spaces.
 - It involves operations on **matrices and vectors**, solving **systems of linear equations**, and understanding geometric concepts like **lines, planes, and subspaces**. – “chatgpt.”



Fig: What is Linear Algebra?

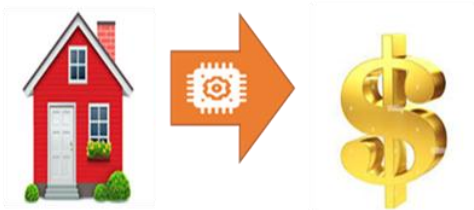
Image: somewhere from web compiled by siman

A.2 Why Linear Algebra for Machine Learning?

• Representation of Data:

- In machine learning, data is typically **represented** as **vectors** and **matrices**. For example, a dataset might be **stored as a matrix** where each row is a data point (vector), and each column is a feature.

Task: House Price Prediction.



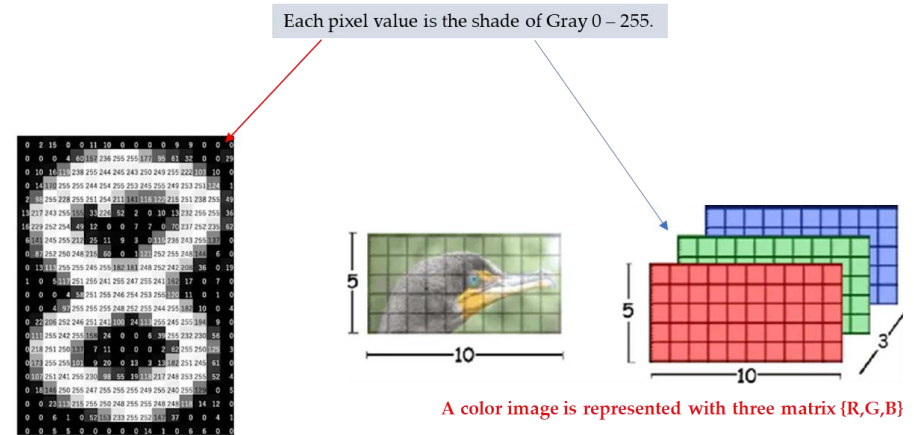
Data: Features/Descriptor of House

Area	Rooms	Price
1080	8	1,00,000.00
1200	10	1,50,000.00

How would you represent this, for computer?

Matrix.

$$\begin{bmatrix} 1080 & 8 \\ 1200 & 10 \end{bmatrix} \begin{bmatrix} 1,00,000 \\ 1,50,000 \end{bmatrix}$$



A gray scale image is represented with single matrix {R,G,B}

A color image is represented with three matrix {R,G,B}

A.2.1 Why Linear Algebra for Machine Learning?

- **Efficient Computing:**

- Matrix operations allow for efficient computations on large datasets. Libraries like **NumPy**, **TensorFlow**, and **PyTorch** leverage **linear algebra** for operations on large matrices and tensors {**Vectorizations**}, which makes **machine learning models faster** and more **scalable**.

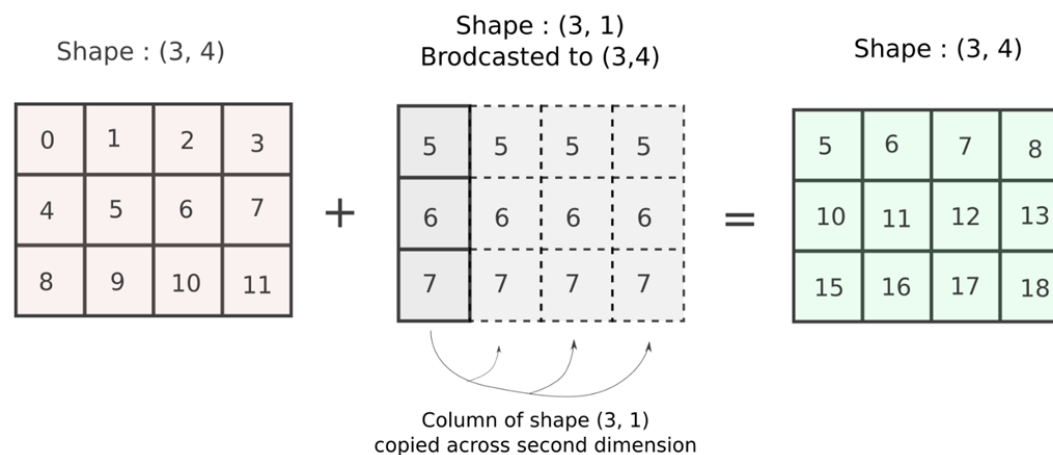
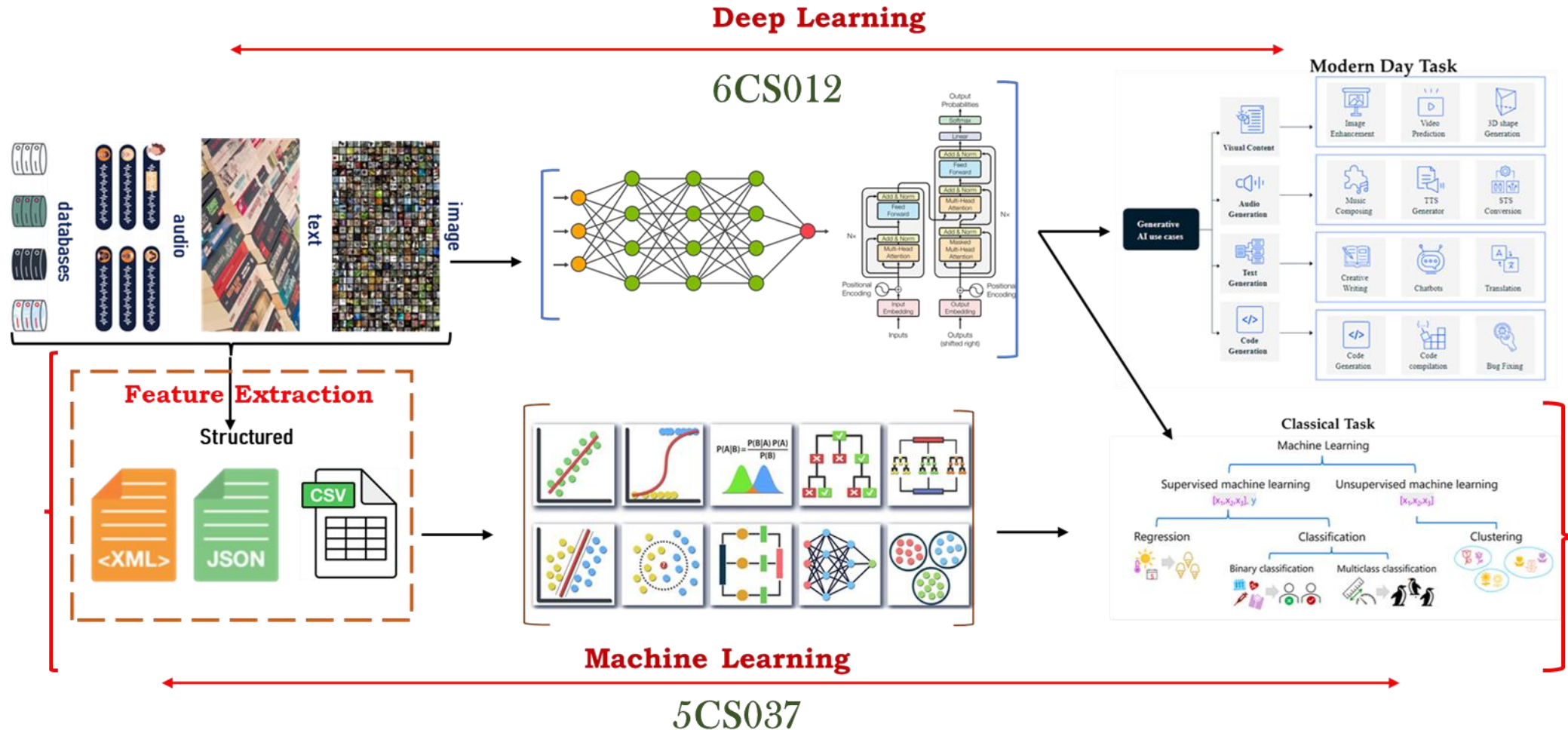


Fig: Idea of Vectorizations.

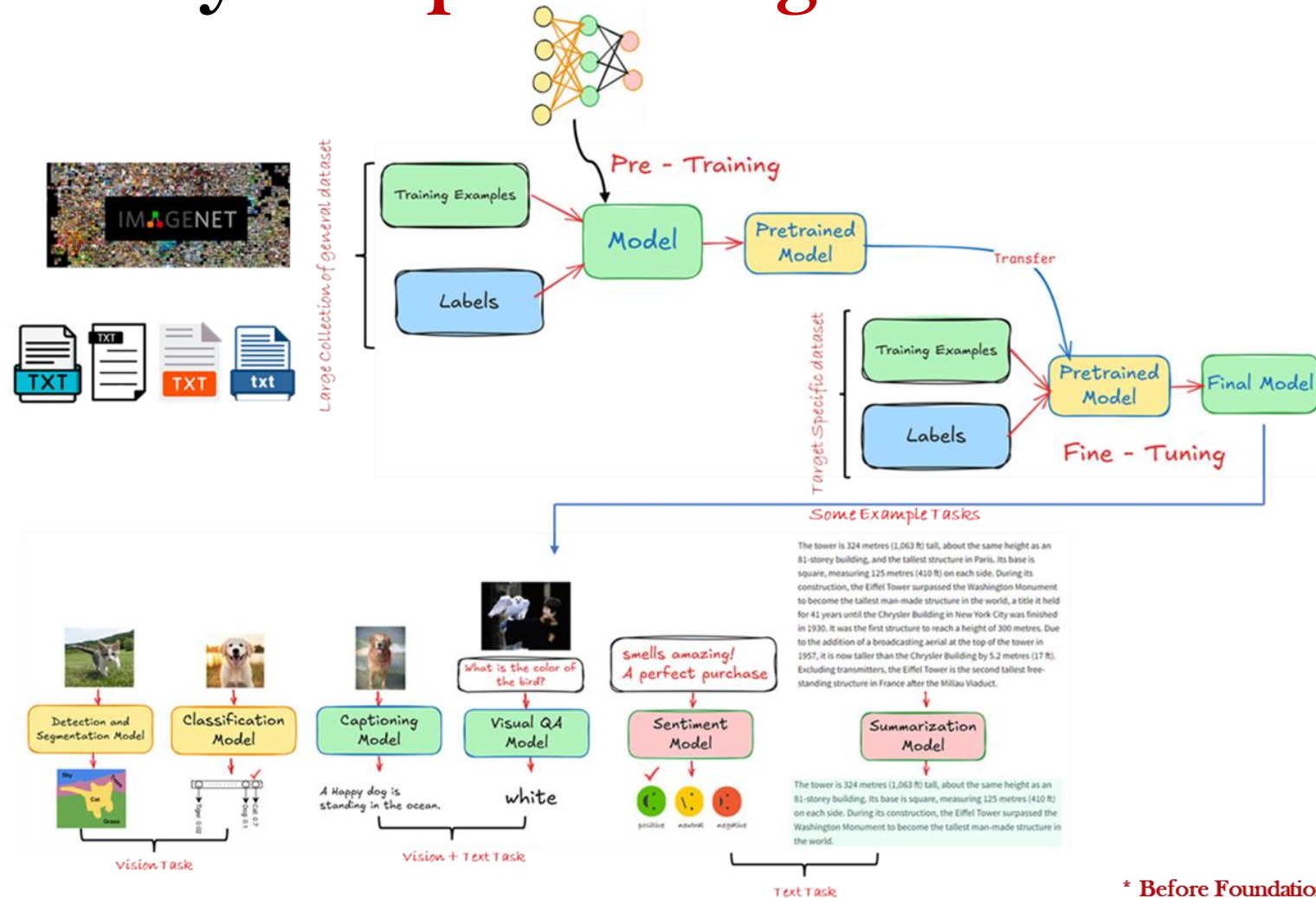
A.2.2 Why Linear Algebra for Machine Learning?

- **Understanding {Machine or Deep Learning} Algorithms:**
 - **Training** machine or deep learning models often involves **solving systems of linear equations**.
 - **Linear algebra** provides the **necessary tools** to solve these systems efficiently.
 - Many machine learning algorithms are based on linear algebra concepts.
 - For instance:
 - **Linear Regression** involves finding a line (or hyperplane) that best fits the data.
 - **Neural Networks** use matrix multiplication for forward and backward propagation.

B. Machine Learning Vs. Deep Learning



C. Summary: Deep Learning Work Flow.



* Before Foundation Models.

Understanding **Vector** and **Matrices**.

{Basic Concepts, Definition and Notations.}

1.1 What are Vectors?

Interpretation – 1: Point in Space.

- E.g., in **2D{dimension}**
 - we can visualize the data points with respect to a **coordinate origin**

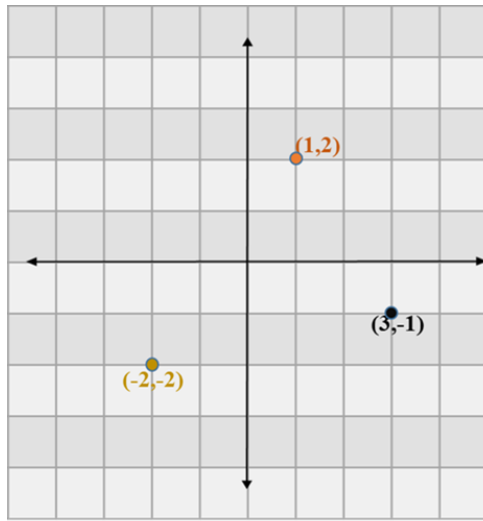


Fig: Vector as a point

Interpretation – 2: Direction in Space.

- E.g., the vector $\vec{v} = [3, 2]^T$ has a **direction** of 3 steps to the right and 2 steps up
- The notation $\vec{v}$ is sometimes used to indicate that the vectors have a **direction**
- All vectors in the figure have the **same direction**

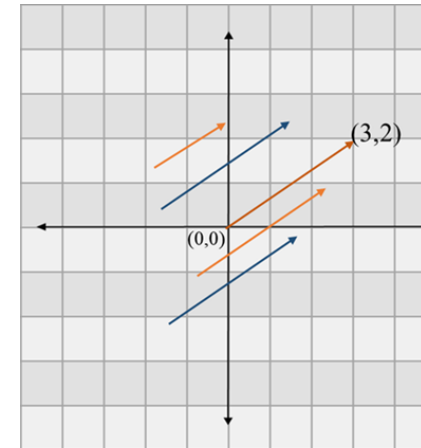


Fig: Vector as Direction

1.2 Vector formal Definition.

- In Linear Algebra and Applied Mathematics, we define vector with in **n-dimensional vector space**.
- **Vector Space**:
 - If **n** is a positive integer, then an **ordered n-tuple** is a sequence of **n real numbers** $[n_1, n_2, \dots, n_n]$
 - The set of all **ordered n-tuples** is called **n – space** or **n – dimensional vector space** and is denoted by $\mathbb{R}^n$.
- **Vectors in $\mathbb{R}^n$** :
 - Let $\mathbb{R}^n = \{(\mathbf{x}_1, \dots, \mathbf{x}_n) : \mathbf{x}_j \in \mathbb{R} \text{ for } j = 1, \dots, n\}$. Then,
 - $\vec{\mathbf{x}} = [\mathbf{x}_1, \dots, \mathbf{x}_n]$ is called **a vector in vector space $\mathbb{R}^n$** .
 - The number $\mathbf{x}_j \rightarrow \mathbf{x}_1, \dots, \mathbf{x}_n$ are called the **components** of $\vec{\mathbf{x}} \in \mathbb{R}^n$.
- Examples:

$$\mathbf{a} = [a_1, a_2] \in \mathbb{R}^2$$

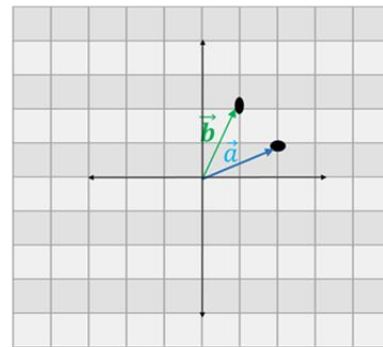


Fig: 2 dimensional vector space

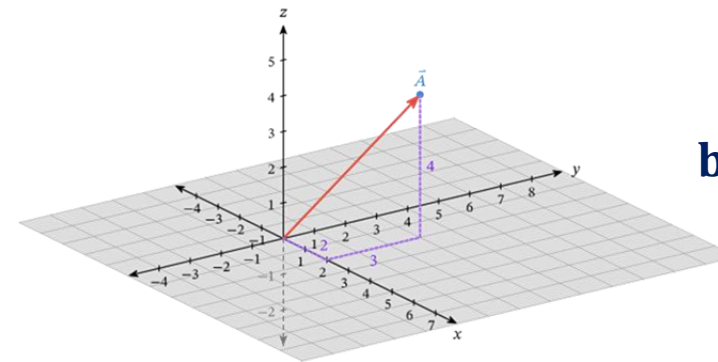


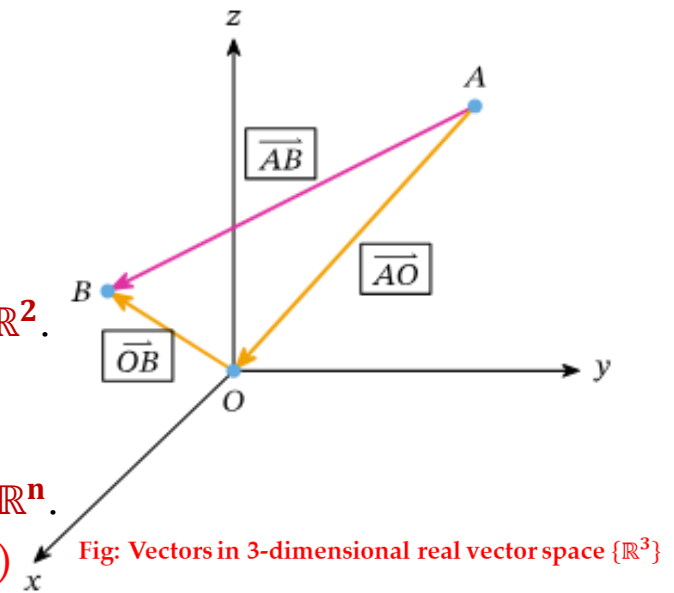
Fig: 3 dimensional vector space

$$\mathbf{b} = [b_1, b_2, b_3] \in \mathbb{R}^3$$

1.3 Vector in Vector – Space.

- **Vector Space:**

- A set **V** of **n -dimensional vectors** (with a corresponding **set of scalars**) such that the **set of vectors** is:
 - “closed” under vector addition.
 - “closed” under scalar multiplication.
 - Origins are defined and fixed {0 vector must exist}
- In other words:
 - **For addition of two vectors:**
 - takes **two vectors** $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$, and it produces the **third vector** $\mathbf{u} + \mathbf{v} \in \mathbb{R}^2$.
 - (addition of vectors – gives another vector in the same set)
 - **For scalar Multiplication:**
 - Takes **a scalar** $\mathbf{c} \in \mathbf{F}$ and a vector $\mathbf{v} \in \mathbb{R}^n$ produces a **new vector** $\mathbf{cv} \in \mathbb{R}^n$.
 - (multiplying a vector by a scalar – gives another vector in the same set)



1.4.1 Axioms of Vector – Space.

- If $\mathbf{V}$ is a set of vectors satisfying the above definition of a **vector space**, then it satisfies the following axioms:
 - **Existence of an Additive Identity**: any vector space $\mathbf{V}$ **must have a zero vector**.
 - **Existence of Negative Vector**: for any vector $\mathbf{v}$ in $\mathbf{V}$ its $-\mathbf{v}$ must also be in $\mathbf{V}$.
 - Has **Arithmetic / Algebraic Properties** – We can perform valid mathematical operations. {details in course note}

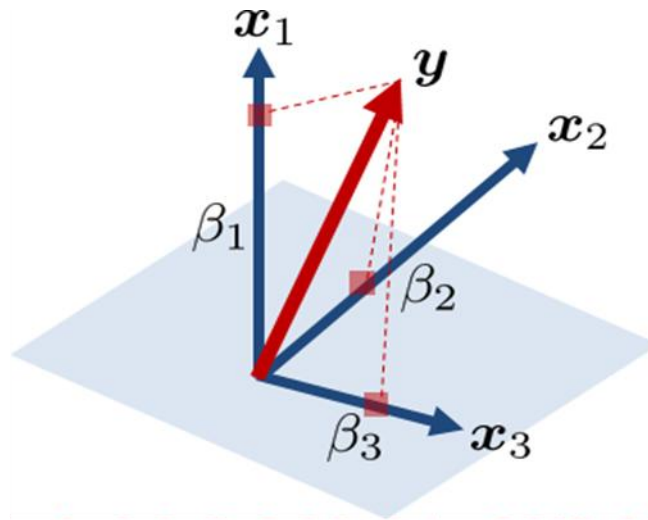


Image from Stanley Chan Book: Introduction to Probability for Data Science.

1.5 Matrices: Introduction.

- In general: A **matrix** is a **rectangular array** of numbers. The **numbers in the array** are called **the entries** in the **matrix**.
 - Array of numbers are an “*ordered collection of vectors*”.
 - Like vectors matrices are also fundamentals in machine learning/AI, as matrices are the way computer *interact with data* in practice.
- A **matrix** is represented with **italicized** upper-case letter like “***A***”.
 - For two dimensions: we say the matrix ***A*** has:
 - m*** rows and ***n*** columns.
 - Each entry/element of ***A*** is defined as ***a_{ij}***.
 - Thus, a **matrix *A^{m×n}*** is define as:
- Overview of notation for discussing matrices:
- Given a set ***C*** $\in \mathbb{R}$, we let ***C_{m×n}*** denote the set of all matrices of ***m*** rows and ***n*** columns consisting of items from set ***C***.
 - For matrix: ***A*** $\in C_{m \times n}$: we let ***a_{ij}*** denote the item at the ***ith*** row and ***jth*** column of ***A***.
 - For matrix ***A*** $\in C_{m \times n}$: we let ***a_{i*}*** denote the ***ith*** row vector of ***A***.
 - For matrix ***A*** $\in C_{m \times n}$: we let ***a_{*i}*** denote the ***jth*** column vector of ***A***.

$$A_{m \times n} := \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}, a_{ij} \in \mathbb{R}$$

1.6 Special Matrices.

- Rectangular Matrix:

- Matrices are said to be rectangular when the number of rows is $\neq$ to the number of columns, i.e. $A^{m \times n}$ with $m \neq n$. For instance:

$$A_{2 \times 3} := \begin{bmatrix} 1 & 2 & 3 \\ 5 & 5 & 4 \end{bmatrix}$$

- Square Matrix:

- Matrices are said to be square when the number of rows = the number of columns, i.e. $A^{m \times n}$. For instance:

$$A_{2 \times 2} := \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$$

- Diagonal Matrix:

- Square matrices are said to be diagonal when each of its non-diagonal elements is zero, i.e. for
 - $D = (d_{ij})$, we have $\forall i, j \in n \ i \neq j \Rightarrow d_{ij} = 0$.
- For instance:

$$A_{3 \times 3} := \begin{bmatrix} 9 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

- Upper triangular matrix:

- Square matrices are said to be upper triangular when the elements below the main diagonal are zero i.e. For $D = (d_{ij})$, we have $d_{ij} = 0$, for $i > j$. For instance:

$$A_{3 \times 3} := \begin{bmatrix} 9 & 8 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 5 \end{bmatrix}$$

- Lower triangular matrix:

- Square matrices are said to be lower triangular when the elements above the main diagonal are zero . i.e. $D = (d_{ij})$, we have $d_{ij} = 0$, for $i < j$. For instance:

$$A_{3 \times 3} := \begin{bmatrix} 9 & 0 & 0 \\ 8 & 1 & 0 \\ 4 & 2 & 5 \end{bmatrix}$$

- Identity Matrix:

- A diagonal matrix is said to be the identity when the elements along its main diagonal are equal to one. For instance:

$$A_{3 \times 3} := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

1.6.1 Special Matrices.

- **Symmetric Matrix:**

- Square matrices are said to be symmetric if they are equal to their transpose, i.e. $\mathbf{A} = \mathbf{A}^T$. For instance:

$$\mathbf{A}_{3 \times 3} := \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 6 \\ 3 & 6 & 1 \end{bmatrix}$$

- **Scalar Matrix:**

- Diagonal matrices are said to be scalar when all the elements along its main diagonal are equal, i.e. $\mathbf{D} = \alpha \mathbf{I}$. For instance:

$$\mathbf{A}_{3 \times 3} := \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

- **Null or Zero Matrix:**

- Matrices are said to be null or zero matrices when all its elements equal to zero, which is denoted as $\mathbf{0}_{m \times n}$. For instance:

$$\mathbf{A}_{3 \times 3} := \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- **Equal Matrix:**

- Two matrices are said to be equal if
 - $\mathbf{A}(a_{ij}) = \mathbf{B}(b_{ij})$.
- For instance:

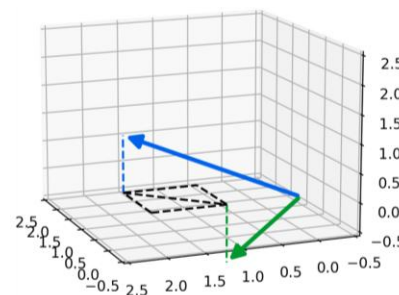
$$\mathbf{B}_{2 \times 2} := \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$$

$$\mathbf{A}_{2 \times 2} := \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$$

1.7 Interpretation of a Matrix: Collection of Vectors.

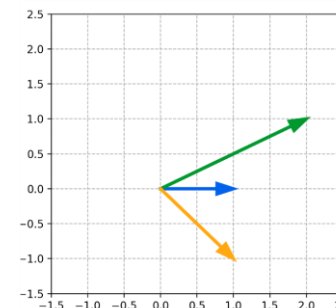
- A matrix can be thought of as a set of vectors.
- For example, for the following matrix:
 - $A := \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$ can be thought of as
 - a **two** three-dimensional **row** vectors i.e.
 - $a_{1*} := [1 \ 2 \ 1]$ and $a_{2*} := [0 \ 1 \ -1]$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$



- Or as a **three** two-dimensional **column** vectors:
- $a_{*1} := \begin{bmatrix} 1 \\ 0 \end{bmatrix}$; $a_{*2} := \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $a_{*3} := \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$



1.7.1 Interpretation of Matrix: As a table of data.

- The simplest interpretation of matrix is as a two – dimensional array of values.
- For example:
 - A numerical dataset represented as a matrix.

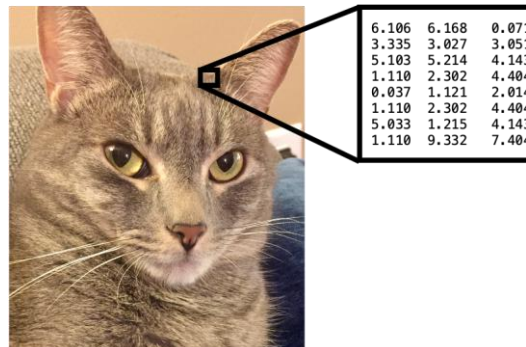
Spreadsheet

	A	B	C	D
1	sepal_length	sepal_width	petal_length	petal_width
2	5.1	3.5	1.4	0.2
3	4.9	3	1.4	0.2
4	7	3.2	4.7	1.4
5	6.5	2.8	4.6	1.5
6	5.8	2.7	5.1	1.9
7	7.1	3	5.9	2.1

Matrix

$$\begin{bmatrix} 5.1 & 3.5 & 1.4 & 0.2 \\ 4.9 & 3.0 & 1.4 & 0.2 \\ 7.0 & 3.2 & 4.7 & 1.4 \\ 6.5 & 2.8 & 4.6 & 1.5 \\ 5.8 & 2.7 & 5.1 & 1.9 \\ 7.1 & 3.0 & 5.9 & 2.1 \end{bmatrix}$$

- The **pixels** of an image can be represented as a **matrix**.
- Let's say we have an image of **$m \times n$ pixels**.
 - Let **X** be a **matrix** representing this image where **$x_{i,j}$** represents the **intensity of the pixel** at row **i** and **j** .



1.7.2 Interpretation of Matrix: As a Function.

- A matrix can also be viewed as a function that maps
 - **vectors in one vector space** to **vectors in another vector space**.
- These special kind of **matrix – defined function** are also called
 - **Linear Transformation** and written as:
 - $T(x) := Ax$
- A very simple visualization of such function is **matrix – vector multiplication**.

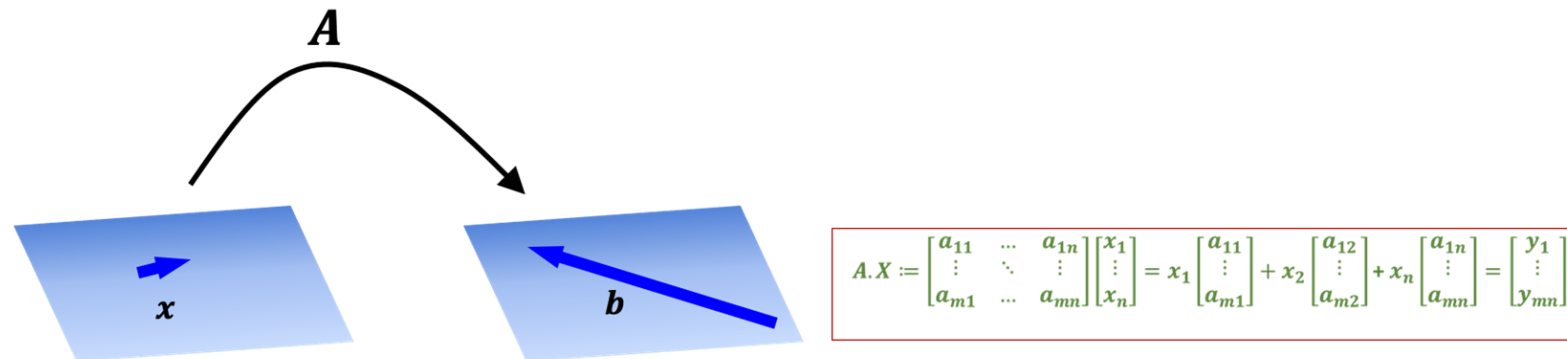
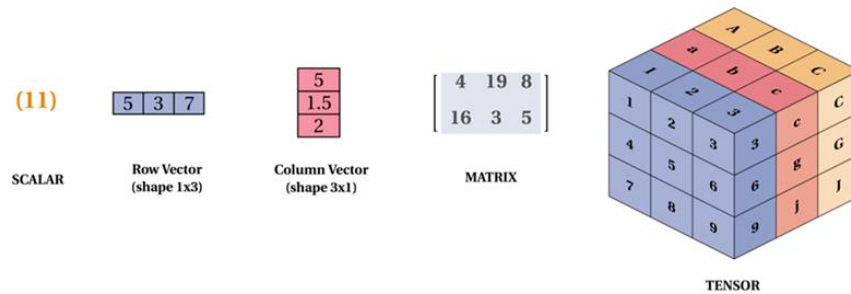


Fig: What happens if we Multiply Matrix A with vector x?

Good to Know!!!

- A tensor is a **multidimensional array** and a **generalization** of the concepts of a **vector** and a **matrix**.



- Tensors can have many axes, here is a tensor with three axes:

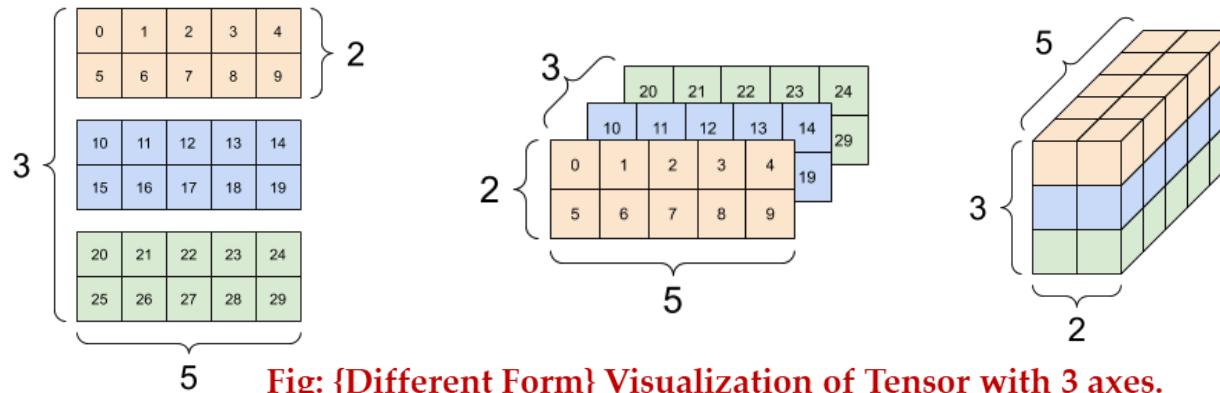
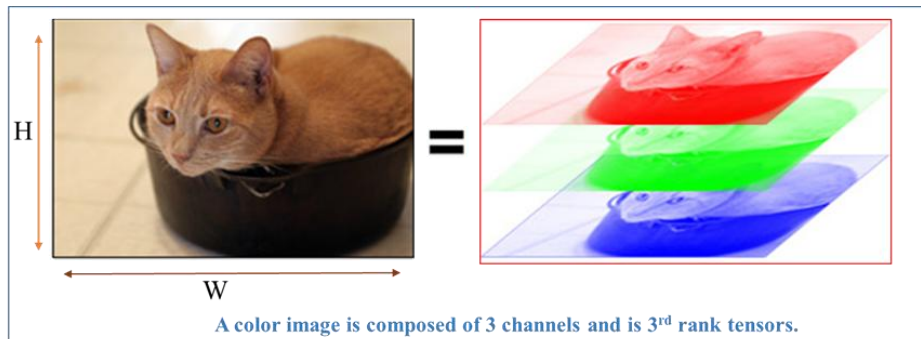


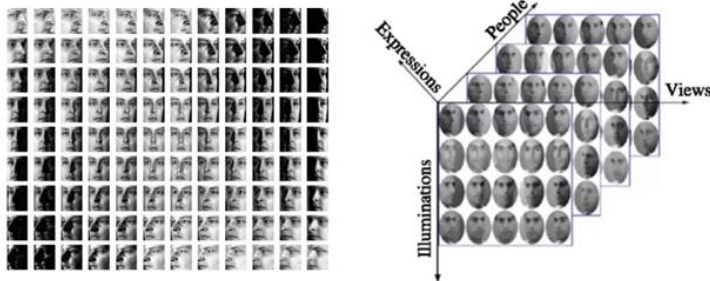
Fig: {Different Form} Visualization of Tensor with 3 axes.

Tensor $\rightarrow$ Example.

- Tensors in DL are Used to represent an image.
 - $\text{image_shape} := \text{Height} \times \text{Width} \times \text{Color Channel (RGB)}$



facial images database is 6th-order tensor



color video is 4th-order tensor



2. The Geometry of Vectors.

{Operations, Linear Dependence, and Basis}

2.1 Understanding Dot Products.

- **Dot product:**

- Given two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, the quantity $\mathbf{u}^T \mathbf{v}$, sometimes called the **inner product** or **dot product** of the vectors, is a real number given by:

- $\mathbf{u}^T \mathbf{v} \in \mathbb{R} = [\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n] \cdot \begin{bmatrix} \mathbf{v}_1 \\ \mathbf{v}_2 \\ \dots \\ \mathbf{v}_n \end{bmatrix} = \sum_{i=1}^n \mathbf{u}_i \times \mathbf{v}_i$

- **Orthogonal Vectors:**

- A pair of vectors $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** if their **dot product** is zero
 - i.e. $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.
- Notation for a pair of orthogonal vectors is $\mathbf{u} \perp \mathbf{v}$ {i.e. **Vector are perpendicular to each other**}.
- In the $\mathbb{R}^n$; this is equal to pair of vector forming a **90°** angle.

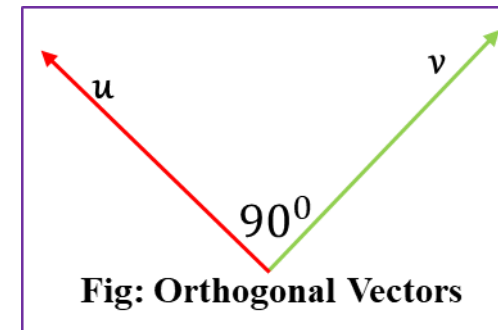
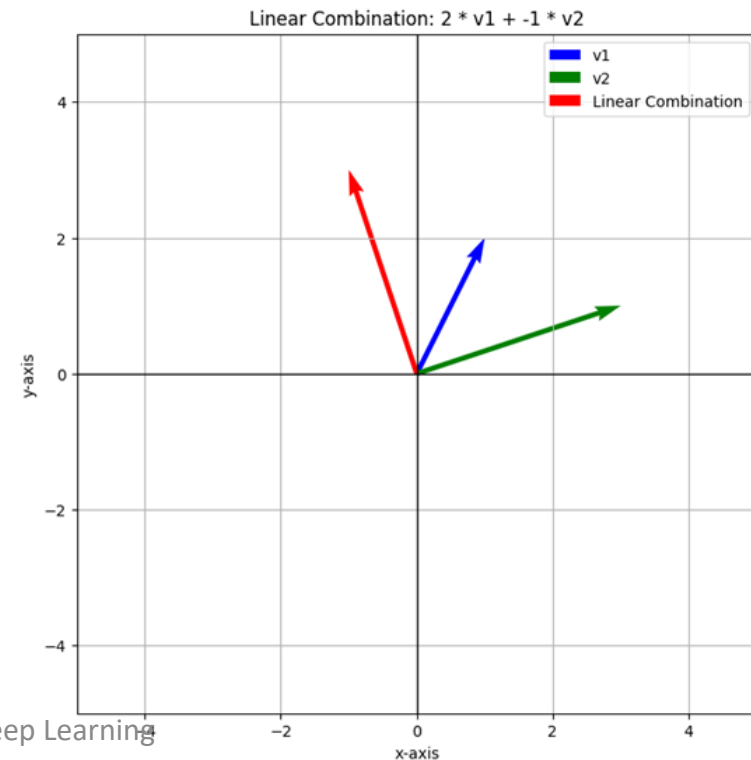


Fig: Orthogonal Vectors

2.2 Linear Combinations of Vectors.

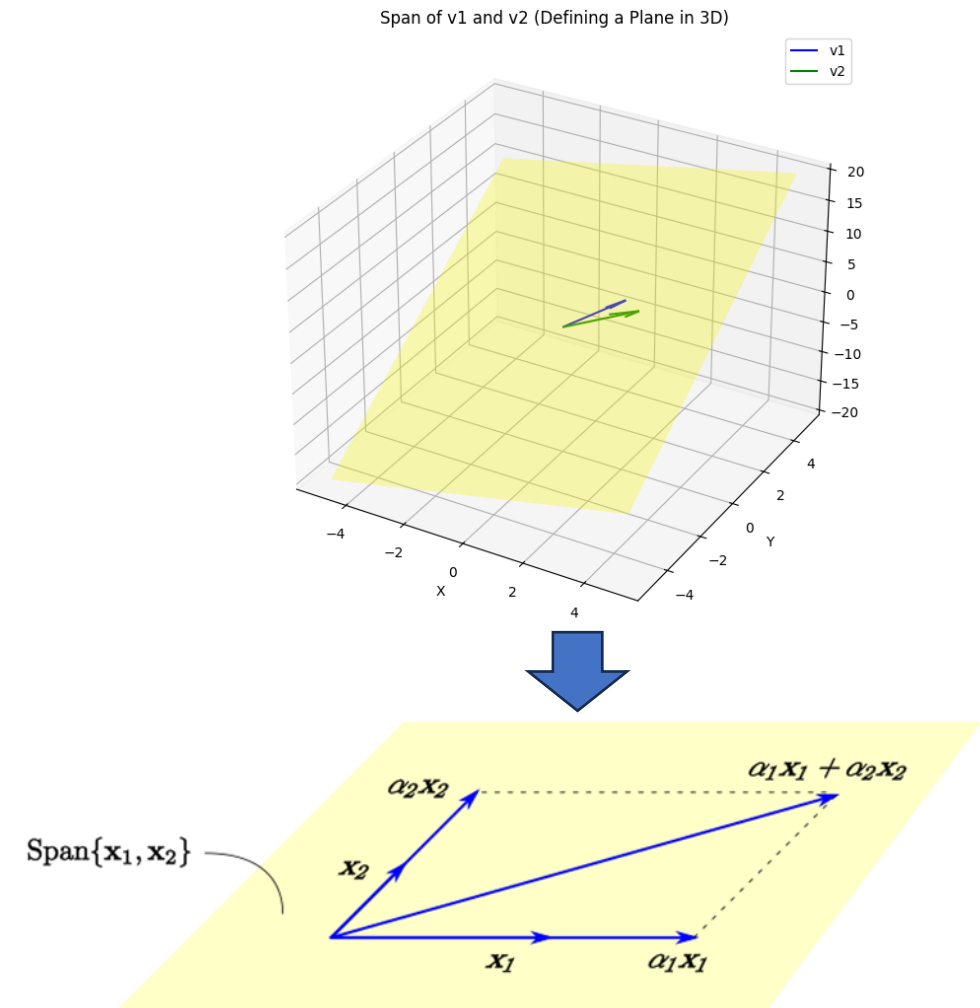
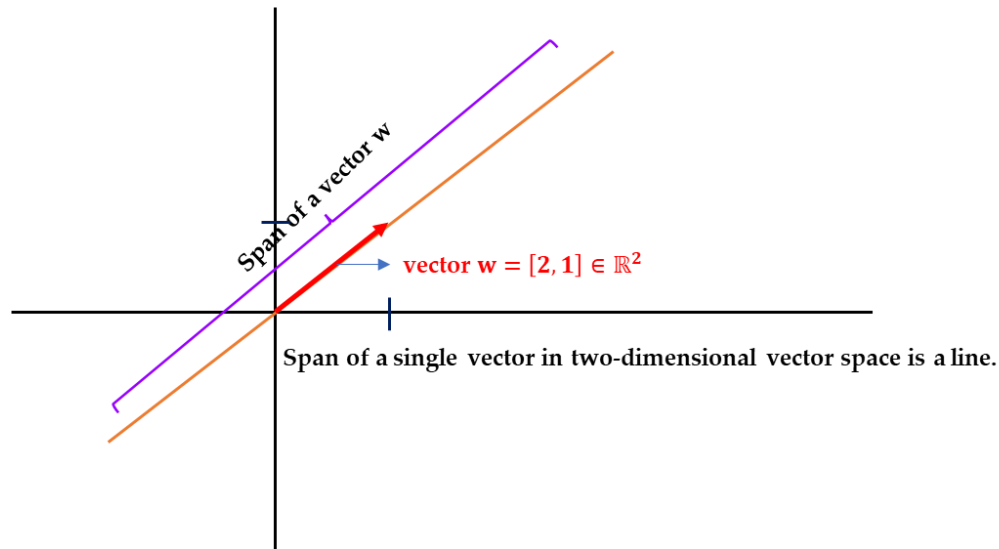
- Idea Combining two or more than vectors to form a new vector.
- Definition:
 - A **vector** $\mathbf{v}$ is a **linear combination** of a set of **vectors** $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, if it can be expressed as:
 - $\mathbf{v} = \mathbf{c}_1\mathbf{v}_1 + \mathbf{c}_2\mathbf{v}_2 + \dots + \mathbf{c}_n\mathbf{v}_n$
 - where:
 - $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$ are scalars (**coefficients**).
 - $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are vectors in a vector space.
- Example in $\mathbb{R}^2$:
 - Let $\mathbf{v}_1 = [1, 2]$ and $\mathbf{v}_2 = [3, 1]$,
 - If we take scalars $\mathbf{c}_1 = 2$ and $\mathbf{c}_2 = -1$,
 - then their **linear combination** will
 - produce a new **vector** $\mathbf{v}$ in same **vector space**.
 - $\mathbf{v} = 2 \times \begin{bmatrix} 1 \\ 2 \end{bmatrix} + (-1) \times \begin{bmatrix} 3 \\ 1 \end{bmatrix}$ ■



2.3 Span of a Set of vectors.

- **Span** is a consequences of **Linear combination of vectors** and can be thought as a **subset** inside a **vector space** (also known as **vector subspace**).
- A subspace, $\mathbb{S}$ of real **vector space** $\mathbb{R}^n$ is thought of a **flat surface (having no curvature) surface** with in $\mathbb{R}^n$:
 - is a collection of **all the vectors in** $\mathbb{S}$ which satisfies the following (algebraic) conditions:
 - The *origin (0 vector)* is contained in $\mathbb{S}$.
 - If vector v_1 and v_2 are in $\mathbb{S}$; then $v_1 + v_2 \in \mathbb{S}$.
 - If $v_1 \in \mathbb{S}$ and α a scalar then $\alpha v_1 \in \mathbb{S}$.
- The **span of a set of vectors** $\{v_1, v_2, \dots, v_n\} \in \mathbb{R}^n$ is the **set** of all **possible linear combinations** of those vectors. Formally, the span of $\{v_1, v_2, \dots, v_n\}$ is:
 - $\text{span}(v_1, v_2, \dots, v_n) = \{c_1 v_1 + c_2 v_2 + \dots + c_n v_n | c_1, c_2, \dots, c_n \in \mathbb{R}\}$
 - where $c_1, c_2, \dots, c_n$ are **scalar coefficients**.

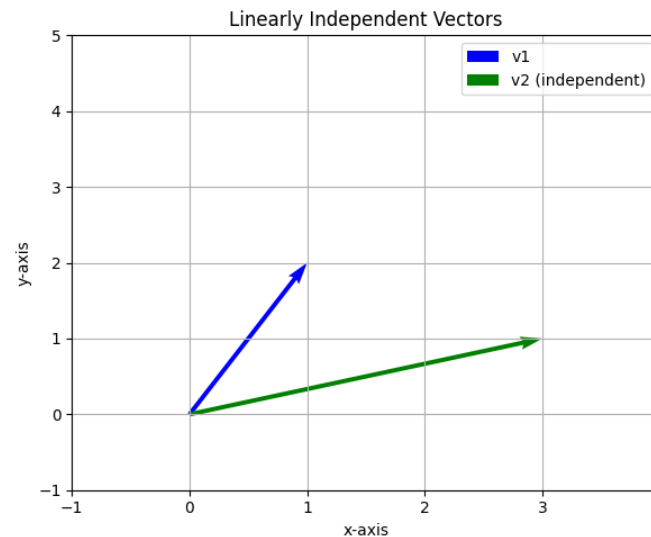
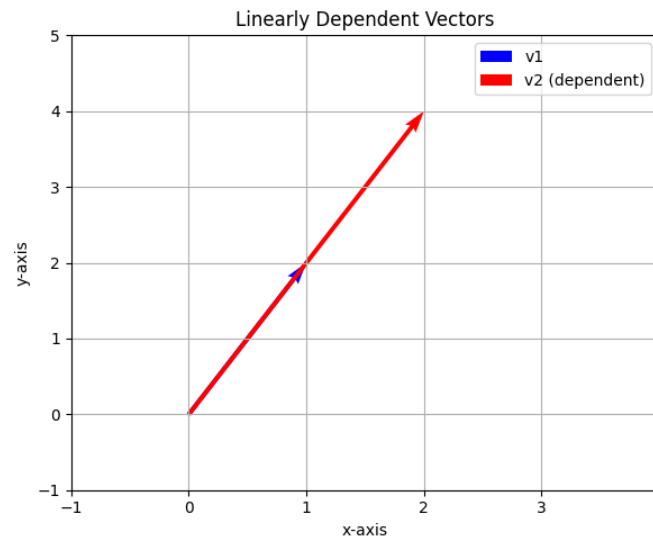
2.3.1 Geometric Interpretation of a Span.



Span of a two vector in three-dimensional space is a plane.

2.4 Linearly Independent and Dependent Vectors.

- A set of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in a **vector space** $\mathbb{R}^n$ is:
 - **Linearly dependent** if at least one vector can be written as a linear combination of the others.
 - Mathematically, this means there exists at least one scalars $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_n$ which is not zero, such that:
 - $\mathbf{c}_1\mathbf{v}_1 + \mathbf{c}_2\mathbf{v}_2 + \dots + \mathbf{c}_n\mathbf{v}_n = \mathbf{0}$; **at least one $\mathbf{c} \neq 0$.**
 - **Linearly Independent** if the only possible solution for above equation is $\mathbf{c}_1 = \mathbf{c}_2 = \dots = \mathbf{c}_n = \mathbf{0}$, i.e. no vector set can be written as a combination of the others.



3. Matrix Algebra.

{Important Matrix Operations.}

3.1 Matrix Determinant.

- **Determinant** of a matrix, denoted by **$\det(\mathbf{A})$ or $|\mathbf{A}|$** , is a **real-valued scalar** encoding certain properties of the matrix

- E.g., for a matrix of size 2×2 :

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

- For larger-size matrices the determinant of a matrix is calculated as

$$\det(\mathbf{A}) = \sum_j a_{ij} (-1)^{i+j} \det(\mathbf{A}_{(i,j)})$$

- In the above, $\mathbf{A}_{(i,j)}$ is a **minor** of the matrix.

- Properties:

- $\det(\mathbf{AB}) = \det(\mathbf{BA})$
- $\det(\mathbf{A}^{-1}) = \frac{1}{\det(\mathbf{A})}$
- $\det(\mathbf{A}) = 0 \rightarrow \mathbf{A}$ is singular i. e. non square matrix.

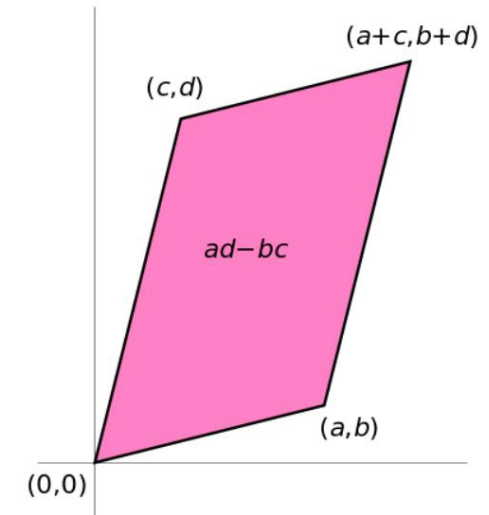
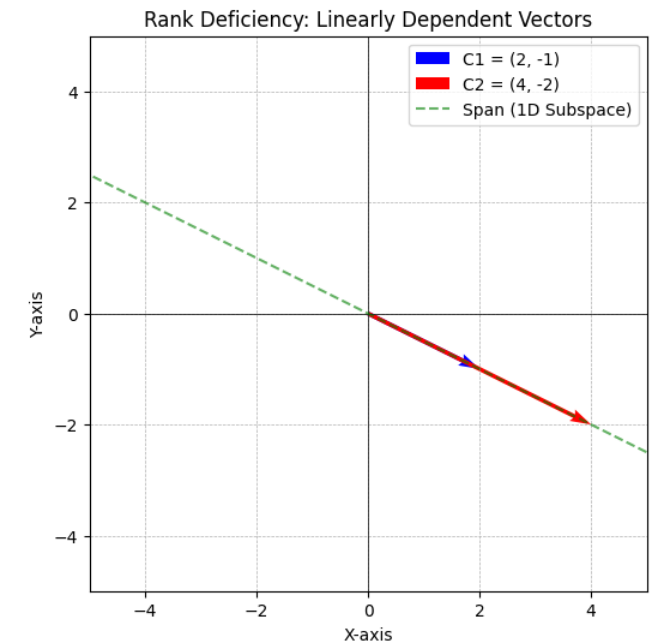
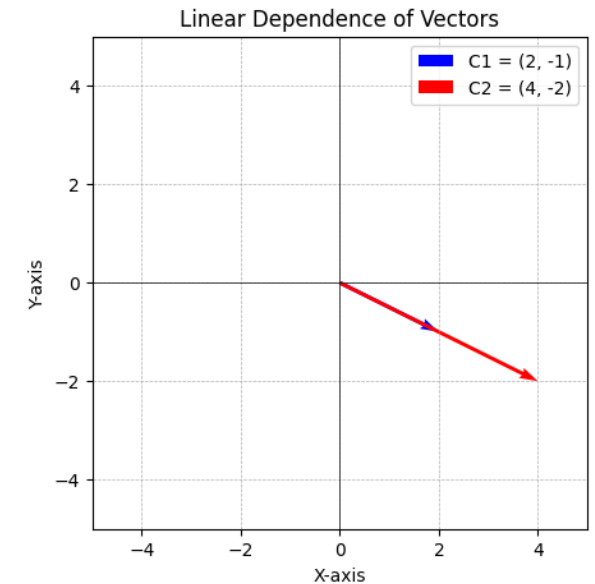


Fig: determinant represents area (or volume) of the parallelogram described by the vectors in the rows of the matrix

3.2 Rank of a Matrix.

- For $m \times n$ matrix the rank of the matrix is the largest number of linearly independent row or columns.
- For Example:
 - For Matrix; $\mathbf{B} := \begin{bmatrix} 2 & -1 \\ 4 & -2 \end{bmatrix}$ Find the Rank and Interpret.
 - Our Observation:**
 - The second column c_2 can be written as : $c_2 = 2 \times c_1$
 - Since one column can be expressed as a multiple of the other, there is only **one independent column**.
 - Thus, the **rank of B is 1**, meaning it can span only a **1 D space in $\mathbb{R}^2$** vector space.
 - Since the **full rank of 2×2 matrix in $\mathbb{R}^2$ vector space is 2**,
 - B is considered **rank – deficient**.



3.3 Inverse of a Matrix.

- The inverse of a square matrix A , denoted as A^{-1} , is a matrix that satisfies:
 - $AA^{-1} = A^{-1}A = I$
 - here I is the identity matrix.
- **Conditions for Invertibility:**
 - A **matrix** $A_{m \times n}$ has an **inverse** if and only if :
 - It is a **square matrix** ($n \times n$).
 - Its **determinant** is **nonzero** i.e. $\det(A) \neq 0$.
 - Its **rank is full**, meaning $\text{rank}(A) = n$.
 - If any of these conditions fail, **the matrix is singular and does not have an inverse**.

3.3.1 Finding inverse of a Matrix.

- If Inverse Exist, we can find the inverse of a Matrix by:
 - **Using Row Reduction:**
 - Row reduction is a method of transforming a matrix into a simpler form (**row echelon form – REF**) usually the identity matrix for finding the inverse.
 - **REF** can be reached via following valid row operations:
 - Swap two rows
 - Multiply a row by a non zero scalar
 - Add or subtract multiples one to/from another row.
 - It can be done using:
 - Gaussian Elimination: **Transform** the **matrix A** into **REF** and then use **back – substitution** to solve for the inverse.
 - Gauss – Jordan Elimination: **Transform** the **matrix A** into the **identity matrix** directly, with **no need for back substitution**.
 - **Using Adjoint (Cofactor) Formula:**
 - Find the inverse of A using the **adjoint** (also called adjugate) **of the matrix**.
 - $A^{-1} = \frac{1}{\det(A)} \times \text{adj}(A),$
 - For 2×2 matrix:
 - $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix};$
 - $A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

3.4 System of Linear Equation.

- A **system of linear equation** is a **collection of one or more linear equations** that share a common set of variables. For example:

- $2x + y = 5$
 $3x + 4y = 6$

- **Types of Systems:**

- **Consistent System:** A system that has at least one solution.
 1. **Unique Solution:** Occurs when the system has a single solution.
 2. **Infinite Solutions:** Occurs when the system has many solutions.
- **Inconsistent System:** A system that has no solution.

3.4.1 Solving System of Linear Equations.

- There are different techniques , our interest is **Matrix Method (aka Matrix Inversion Method)**.
 - Any system of Linear Equation:
 - $a_{11}x_1 + a_{12}x_2 = b_1$
 $a_{21}x_1 + a_{22}x_2 = b_2$
 - can be represented in the form:
 - $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \cdot \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ i. e. $\rightarrow Ax = b$
 - here:
 - $A \rightarrow$ is a matrix of coefficients with size $m \times n$, m is the number of equations and n is the number of variable.
 - $x \rightarrow$ is a column vector representing the unknown variables with size $n \times 1$.
 - $b \rightarrow$ is a column vector representing the constants with size $m \times 1$.
 - The equation can be modified:
 - $A^{-1}Ax = A^{-1}b$ {Multiplying both side by A^{-1} }
 - $Ix = A^{-1}b$ {I is the identity matrix}
 - $x = A^{-1}b$ {you know how to find A^{-1} }

3.5 Matrix – Matrix Multiplication.

- Matrix multiplication between $\mathbf{A} \in \mathbb{R}^{n \times p}$ and $\mathbf{B} \in \mathbb{R}^{p \times q}$ with resultant matrix $\mathbf{C} \in \mathbb{R}^{n \times q}$ can be defined as:

$$\mathbf{A} \cdot \mathbf{B} := \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & \dots & b_{1p} \\ \vdots & \ddots & \vdots \\ b_{n1} & \dots & b_{np} \end{bmatrix} = \begin{bmatrix} c_{11} & \dots & c_{1p} \\ \vdots & \ddots & \vdots \\ c_{m1} & \dots & c_{mp} \end{bmatrix}$$

$$c_{ij} := \sum_{l=1}^n a_{il} b_{lj}; \text{ with } i=1, \dots, m; \text{ and } j=1, \dots, p$$

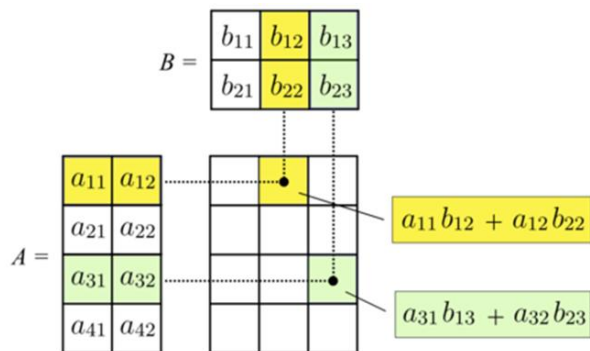


Fig: Schematic representation of Matrix product

Properties of Matrix – Matrix Multiplication:

1. Associativity: $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$
2. Associativity with scalar Multiplication: $\alpha(\mathbf{AB}) = (\alpha\mathbf{A})\mathbf{B}$
3. Distributive with sum: $\mathbf{A}(\mathbf{B} \mp \mathbf{C}) = \mathbf{AB} \mp \mathbf{AC}$
4. Cautions!! In matrix – matrix multiplication orders matter, it is not commutative i.e. $\mathbf{AB} \neq \mathbf{BA}$.

3.6 Matrix – Vector Multiplication.

- Matrix-vector multiplication is an operation between a matrix and a vector that produces a new vector.
- Matrix-vector multiplication equals to taking the dot product of each column n of matrix- A with each element of vector- x resulting in vector y and is defined as:

$$A.X := \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ \vdots \\ a_{m2} \end{bmatrix} + x_n \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} y_1 \\ \vdots \\ y_{mn} \end{bmatrix}$$

- Matrix – vector multiplication can be interpreted as taking a **linear combination** of the **columns** of a matrix A **weighted** by elements of **vector x** .
 - What can be the consequences of such operation?

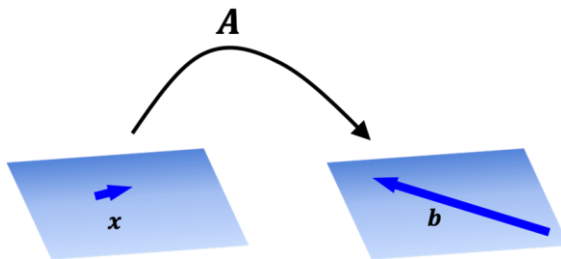


Fig: How my vector will Transformed?

Matrix – vector multiplication can result in:
 Change in magnitude or,
 Change in direction or,
 Both changes depending on the matrix involved.

3.6.1 Geometric Interpretation of Matrix – vector Multiplication.

- **Rotation Matrix:**

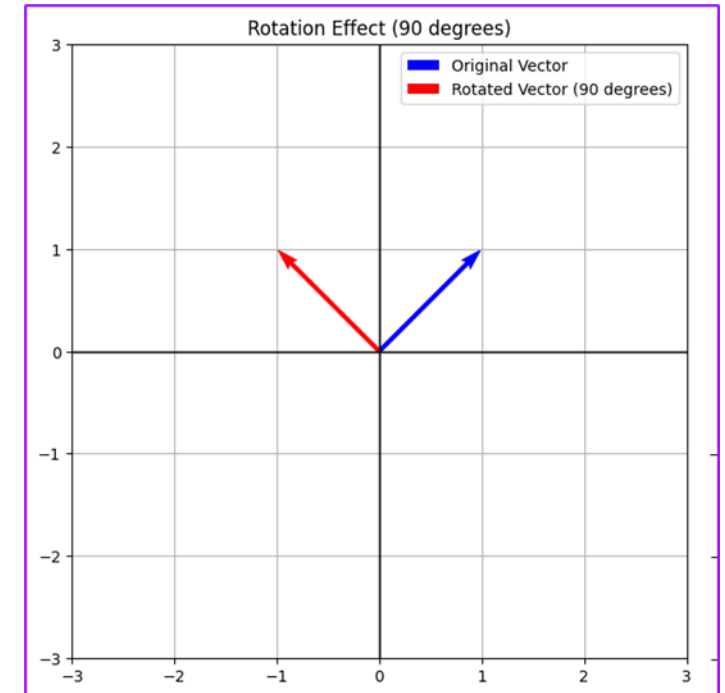
- A rotation matrix rotates a vector by a specified angle while preserving its magnitude.
- **Example:** A 2D rotation matrix that rotates a vector by 90 degrees counterclockwise:

- $\mathbf{R} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

- **Effect:** This matrix rotates the vector without changing its length.

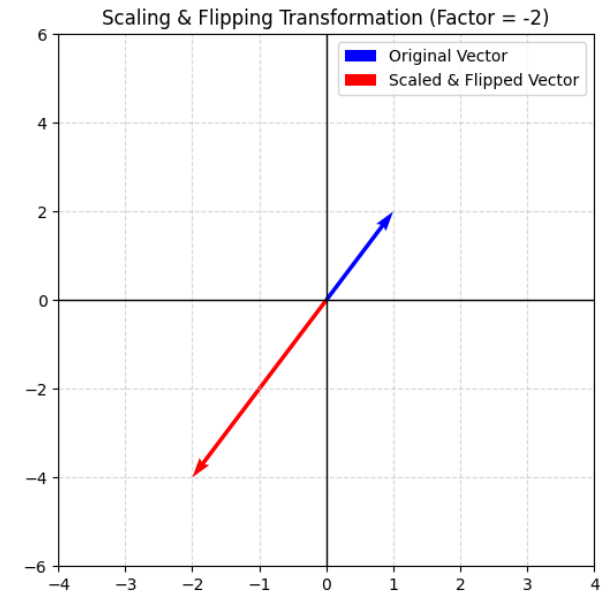
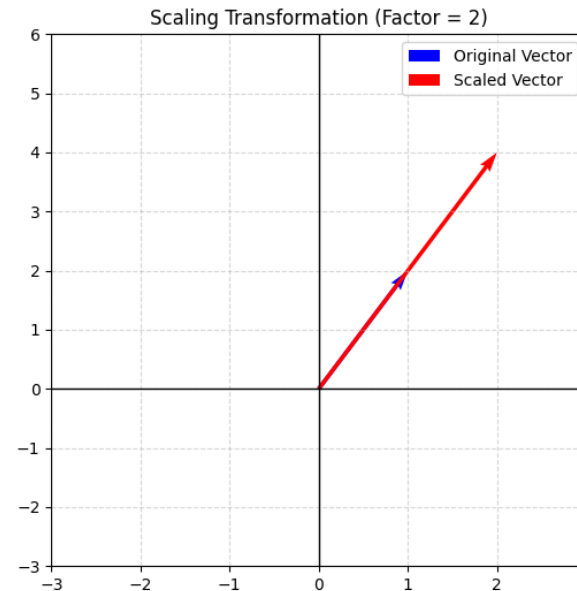
- **Example Calculation:**

- Given the vector $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \rightarrow \mathbf{Rv} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
 - The **magnitude remains 1**, but the **direction changes** from the **x-axis** to the **y-axis**.



3.6.2 Geometric Interpretation of Matrix – vector Multiplication.

- **Scaling Matrix:**
 - A scaling matrix increases or decreases the magnitude of the vector without changing their direction.
 - $S = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$
 - where k is the **scaling factor**:
 - If $k > 1$, the vector is stretched.
 - If $0 < k < 1$, the vector is compressed.
 - If $k < 0$, the vector is **flipped** and scaled.
- **Example:**
 - Given a vector: $v = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and;
 - a scaling matrices
 - i) $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ and ii) $\begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}$;
 - Applying S to v :
 - i) $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$
 - ii) $\begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ -4 \end{bmatrix}$



4. Eigen Value Problem.

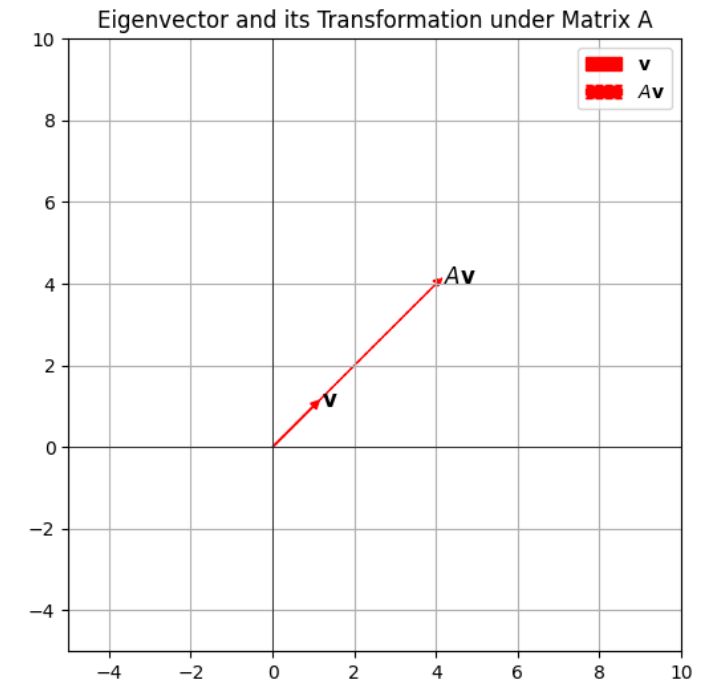
{aka eigen Value Decomposition.}

4.1 Eigen Vector and Eigen Value.

- An **eigenvector** of a **square matrix A** is a **non-zero vector v** that, when **multiplied by A**, results in a scalar multiple of itself.
 - In other words, it is a vector that does not change direction when the linear transformation represented by A is applied to it.
 - Instead, it only gets scaled by a certain factor, called the **eigenvalue**.
- Mathematically, for any **Matrix – vector** pair if following holds:
 - $Av = \lambda v$
 - then the **vector v** is called **eigen vector** and the **scaling factor λ** is called **eigen value**.
- **Key points about eigen vectors:**
 - **Non-zero:** Eigenvectors are always non-zero vectors, **i.e. $v \neq 0$** .
 - **Scaling:** The transformation A simply scales the eigenvector by the **eigenvalue λ** ;
 - it does not change the **vector's direction**.
 - **Multiple eigenvectors:** For each eigenvalue, there can be infinitely many eigenvectors, all scalar multiples of each other. They form a subspace (**called the eigenspace**) corresponding to that eigenvalue.

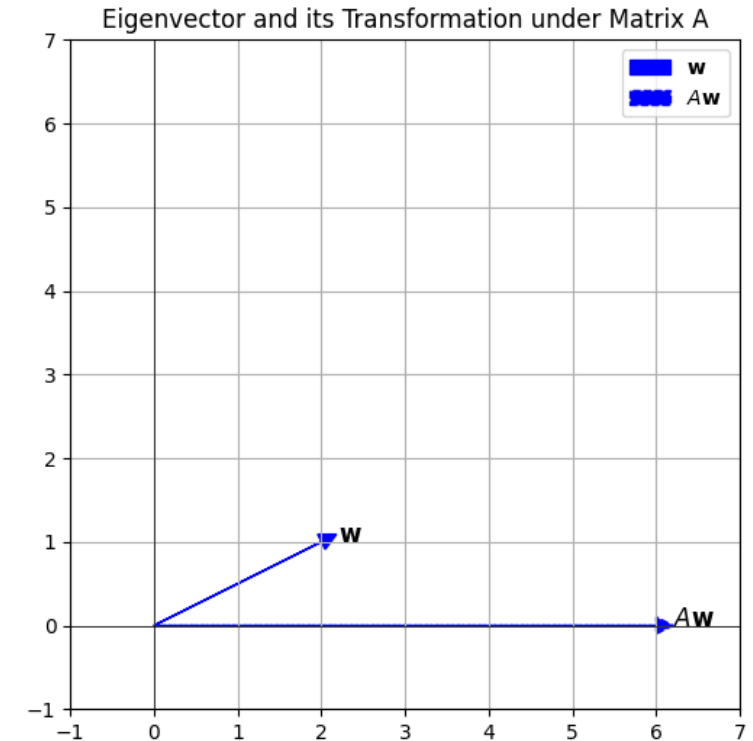
4.1.1 Identify the Eigen Vector.

- Consider a matrix
 - $A = \begin{bmatrix} 2 & 2 \\ -4 & 8 \end{bmatrix}$ and
 - vectors $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$; $\mathbf{w} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.
- Which are Eigen vectors?
 - For $\mathbf{v} \rightarrow$ we check if $\mathbf{v}$ is an eigenvector by calculating $A\mathbf{v}$:
 - $A\mathbf{v} = \begin{bmatrix} 2 & 2 \\ -4 & 8 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \Rightarrow 4\mathbf{v}$.
 - So, $\mathbf{v}$ is an eigen vector with eigenvalue $\lambda = 4$.
- What about $\mathbf{w}$?



4.1.1 Identify the Eigen Vector.

- Consider a matrix
 - $A = \begin{bmatrix} 2 & 2 \\ -4 & 8 \end{bmatrix}$ and
 - vectors $\mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$; $\mathbf{w} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.
- Which are Eigen vectors?
 - For $\mathbf{w} \rightarrow$ we check if $\mathbf{w}$ is an eigenvector by calculating $A\mathbf{w}$:
 - $A\mathbf{w} = \begin{bmatrix} 2 & 2 \\ -4 & 8 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \end{bmatrix} \neq \lambda \mathbf{w}$.
 - So, $\mathbf{w}$ is not an eigen vector there does not exist a
 - scalar λ under which $A\mathbf{w} = \lambda \mathbf{w}$ holds true.



4.2 Eigen Value Problem.

- The **eigenvalue problem** is a fundamental concept in linear algebra and plays a critical role in various fields such as machine learning, physics, and computer science.
- It involves **finding scalar values** (called **eigenvalues**) and corresponding **non-zero vectors** (called **eigenvectors**) for a given **square matrix**.
 - Mathematically, Given a **square matrix A** , the eigenvalue problem is to find **scalars λ** and **eigen vector v** that satisfy the following equation.
 - **$Av = \lambda v$.**

4.3 Steps to solve the Eigenvalue Problem.

- Write the characteristic equation:
 - To find the **eigenvalues**, we rewrite the equation as:
 - $(\mathbf{A} - \lambda \mathbf{I})\mathbf{v} = \mathbf{0}$ {called characteristic equation}
 - Where:
 - $\mathbf{I}$ is the **identity matrix** of the **same size** as $\mathbf{A}$,
 - $\lambda \rightarrow$ **eigen values**.
 - $\mathbf{v} \rightarrow$ **eigen vector**.
 - *Cautions: the matrix $\mathbf{A} - \lambda \mathbf{I}$ must be singular i.e. $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$.*
- Compute the characteristic polynomial:
 - Solve
 - $\det(\mathbf{A} - \lambda \mathbf{I}) = 0$,
 - which gives a **polynomial equation in λ** which is called **characteristic polynomial**.
- Solve the characteristic polynomial:
 - Solve the **polynomial equation** to find the eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$.
- Find the eigen vectors:
 - For each **eigen value λ_i** ,
 - substitute it back into the equation $(\mathbf{A} - \lambda \mathbf{I})\mathbf{v} = \mathbf{0}$ and solve for the **eigenvector $\mathbf{v}$** .

4.3.1 Example Problem.

Eigenvalues of Matrix A

Consider a matrix A :

$$A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$$

The characteristic equation is:

$$\det(A - \lambda I) = \det \begin{pmatrix} 4 - \lambda & 2 \\ 1 & 3 - \lambda \end{pmatrix} = 0$$

Expanding the determinant:

$$(4 - \lambda)(3 - \lambda) - 2 \times 1 = 0$$

Simplifying:

$$\lambda^2 - 7\lambda + 10 = 0$$

Solving this quadratic equation:

$$\lambda_1 = 5, \quad \lambda_2 = 2$$

4.3.1 Example Problem.

Eigenvectors of Matrix A

Next, we find the eigenvectors:

For $\lambda_1 = 5$, solve $(A - 5I)v = 0$:

$$(A - 5I) = \begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix}$$

Solving the system:

$$\begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0$$

This gives the eigenvector:

$$v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

For $\lambda_2 = 2$, solve $(A - 2I)v = 0$:

$$(A - 2I) = \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix}$$

Solving the system:

$$\begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0$$

This gives the eigenvector:

$$v_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

Conclusion: For the matrix A , the eigenvalues are $\lambda_1 = 5$ and $\lambda_2 = 2$, with corresponding eigenvectors $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

and $v_2 = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$.

4.4 Eigenvalue Decomposition.

- **Eigenvalue Decomposition** is a process where a **square matrix is factorized** into
 - its **eigenvalues** and **eigenvectors**.
 - Specifically, for a matrix A , if it can be decomposed into a product of three matrices:
 - $A = V\Lambda V^{-1}$
 - where:
 - A is the original matrix.
 - V is the matrix whose columns are the eigenvectors of A .
 - Λ is a diagonal matrix whose diagonal entries are the eigenvalues of A .
 - V^{-1} is the inverse of the matrix V .
- One of the application of Eigenvalue decomposition is **Principal Component Analysis** used for dimensionality reduction purposes.
 - {This workshop we will implement PCA with eigen value decomposition and try to compress the image.}

Plan

- Tutorial:
 - Review of Derivative for Deep Learning.
 - Some Hands-on Exercise on Vector, Matrices.
- Workshop – Implement PCA with eigenvalue decomposition for Image Compression Application.

The – End.